

Statistical Methods in Chemistry

Block 4

Statistical Comparison and Hypothesis Testing

Student handout

Name: _____

Class: _____

Date: _____

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1 Why chemists compare results

After calculating a mean, standard deviation, and confidence interval, a chemist often has to make a decision. A measured value may look different from an accepted value, or two methods may give different means. The statistical question is whether the observed difference is large enough compared with the experimental scatter.

Core idea

A statistical comparison asks whether an observed difference is larger than we would reasonably expect from random variation alone.

Common chemical comparison questions include:

Question type	Chemistry example
Measured value vs accepted value	Is the measured concentration of a standard significantly different from the certified value?
Two independent groups	Do two analytical methods give different concentrations for the same analyte?
Before–after / paired samples	Did purification significantly reduce the concentration of an impurity in the same samples?
Trend or relationship	Is the slope of a calibration curve significantly different from zero?

A difference is not automatically meaningful

A numerical difference can be caused by random scatter. A statistically significant difference can also be too small to matter chemically. You must consider both statistics and chemistry.

Quick exercise: Recognising comparison questions

For each case, state what is being compared.

- A student measures a sodium carbonate standard and compares the result with $0.1000 \text{ mol dm}^{-3}$.
- A class compares calcium concentration measured by EDTA titration and by an ion-selective electrode.
- The same water samples are tested before and after a filtration step.

2 Null hypothesis, alternative hypothesis, and significance level

A hypothesis test begins by setting up two competing explanations.

Hypotheses

The **null hypothesis**, H_0 , usually says that there is no real difference, no change, or no effect. The **alternative hypothesis**, H_1 , says that there is a real difference, change, or effect.

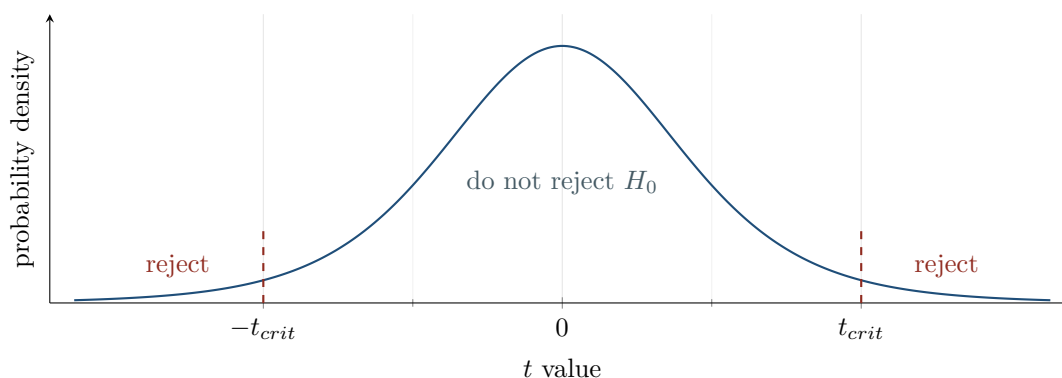
Examples:

Situation	H_0	H_1
One sample vs accepted value	Mean equals accepted value.	Mean differs from accepted value.
Two methods	Method A and Method B have the same mean.	The two method means differ.
Before–after treatment	Mean difference is zero.	Mean difference is not zero.

The **significance level**, usually written α , is the risk of falsely rejecting H_0 that you are willing to accept. A common value is

$$\alpha = 0.05.$$

At the 5 % level, we reject H_0 only if the observed result would be quite unlikely under the assumption that H_0 is true.



What a p-value is not

A p-value is not the probability that H_0 is true. It is the probability of observing data at least this extreme *if* H_0 were true.

Quick exercise: Writing hypotheses

A student tests whether a vitamin C tablet contains the labelled value of 500 mg. Write H_0 and H_1 for a two-sided test.

3 The one-sample t-test

Use a one-sample t-test when a set of repeated measurements is compared with a known or accepted value. This is common when checking a standard solution, certified reference material, or labelled concentration.

One-sample t-test

For n repeated measurements with mean $\bar{x}$ and sample standard deviation s , compare the mean with an accepted value μ_0 using

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

The degrees of freedom are

$$df = n - 1.$$

The denominator $s/\sqrt{n}$ is the **standard error of the mean**. It estimates how uncertain the sample mean is.

Worked example: Checking a standard solution

A student standardises a sodium hydroxide solution against a primary standard. The accepted concentration is $0.1000 \text{ mol dm}^{-3}$. Five independent determinations give:

$$0.1012, \quad 0.1008, \quad 0.1016, \quad 0.1011, \quad 0.1015 \text{ mol dm}^{-3}.$$

For these data:

$$\bar{x} = 0.10124, \quad s = 0.000321, \quad n = 5.$$

The test statistic is

$$t = \frac{0.10124 - 0.1000}{0.000321/\sqrt{5}} = 8.64.$$

With $df = 4$, the two-tailed critical value at $\alpha = 0.05$ is approximately 2.78. Since $8.64 > 2.78$, reject H_0 . The measured concentration is statistically higher than the accepted value.

This points to systematic error

If repeated measurements are precise but significantly shifted from the accepted value, the problem is probably not random scatter. Possible causes include a misprepared standard, contaminated reagent, endpoint bias, or incorrect calculation.

Quick exercise: One-sample t-test calculation

A chloride standard has an accepted concentration of 50.0 mg L^{-1} . A method gives the following results:

$$50.7, \quad 50.5, \quad 50.9, \quad 50.6, \quad 50.8 \text{ mg L}^{-1}.$$

Given $\bar{x} = 50.70$ and $s = 0.158$, calculate t and decide whether the method is significantly different from the accepted value at $\alpha = 0.05$. Use $t_{crit} = 2.78$ for $df = 4$.

4 Comparing two independent means

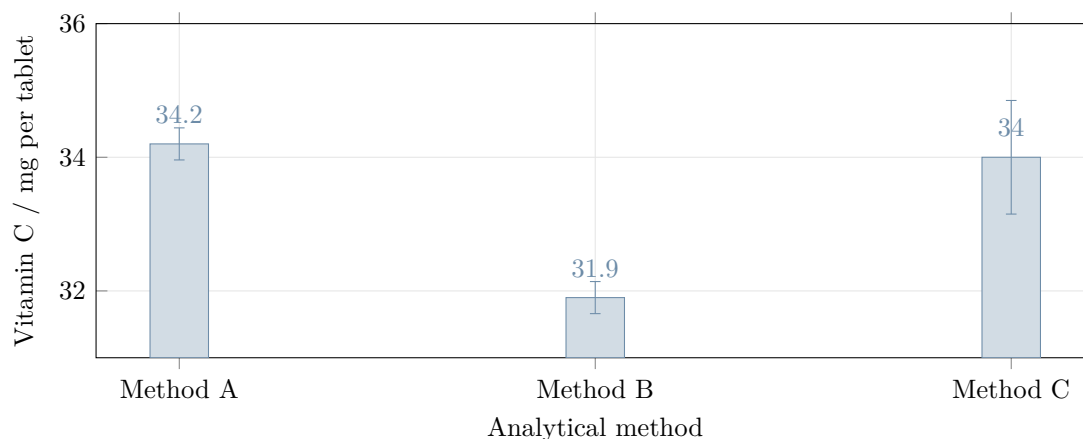
Use a two-sample t-test when comparing two independent groups. For chemical work, a common example is comparing two analytical methods applied to separate replicate samples.

Two-sample comparison

For two independent groups, compare the difference between their means with the combined uncertainty of those means. A useful form is Welch's t-test:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}.$$

Welch's test does not require the two groups to have equal variance, so it is a safe default for school-level method comparison.



The graph shows that Method A and Method B have similar spread, but their means are separated by much more than their typical scatter. Method C has a mean close to Method A, but it is less precise.

Worked example: Comparing two vitamin C methods

Two independent methods are used to estimate vitamin C content. Five tablets are analysed by each method.

	Method A / mg	Method B / mg
Replicate 1	34.1	31.8
Replicate 2	34.5	32.0
Replicate 3	33.9	31.6
Replicate 4	34.2	32.2
Replicate 5	34.4	31.7
Mean	34.22	31.86
Sample s	0.239	0.241

Using Welch's t-test:

$$t = \frac{34.22 - 31.86}{\sqrt{\frac{0.239^2}{5} + \frac{0.241^2}{5}}} = 15.56.$$

This is far larger than a typical 5% two-tailed critical value for this sample size. The two methods give significantly different results.

Quick exercise: Two independent means

Two students measure the same vinegar sample using different procedures.

	Procedure A / mol dm ⁻³	Procedure B / mol dm ⁻³
Mean	0.812	0.835
Sample s	0.006	0.009
n	4	4

Calculate the t-statistic using Welch's formula. Use $t_{crit} \approx 2.45$ as a rough decision boundary. Are the procedures significantly different?

5 Paired t-test: before–after measurements

Sometimes two values are not independent. If the same physical sample is measured before and after treatment, the correct comparison uses the differences within each pair.

Paired t-test

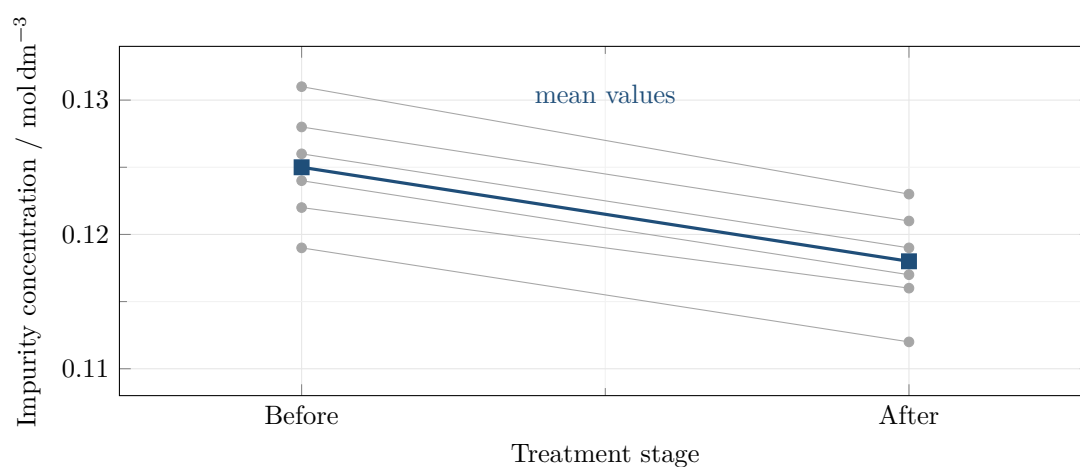
For paired data, first calculate the difference for each pair:

$$d_i = \text{after}_i - \text{before}_i.$$

Then test whether the mean difference $\bar{d}$ is significantly different from zero:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}},$$

where s_d is the sample standard deviation of the differences.



Worked example: Purification step

Six water samples are measured before and after a purification step.

Sample	Before / mol dm ⁻³	After / mol dm ⁻³	Difference d
1	0.124	0.117	-0.007
2	0.131	0.123	-0.008
3	0.128	0.121	-0.007
4	0.119	0.112	-0.007
5	0.122	0.116	-0.006
6	0.126	0.119	-0.007

The mean difference is $\bar{d} = -0.0070$ and the sample standard deviation of the differences is $s_d = 0.00063$. Therefore,

$$t = \frac{-0.0070}{0.00063/\sqrt{6}} = -27.1.$$

The purification step caused a statistically significant decrease in impurity concentration.

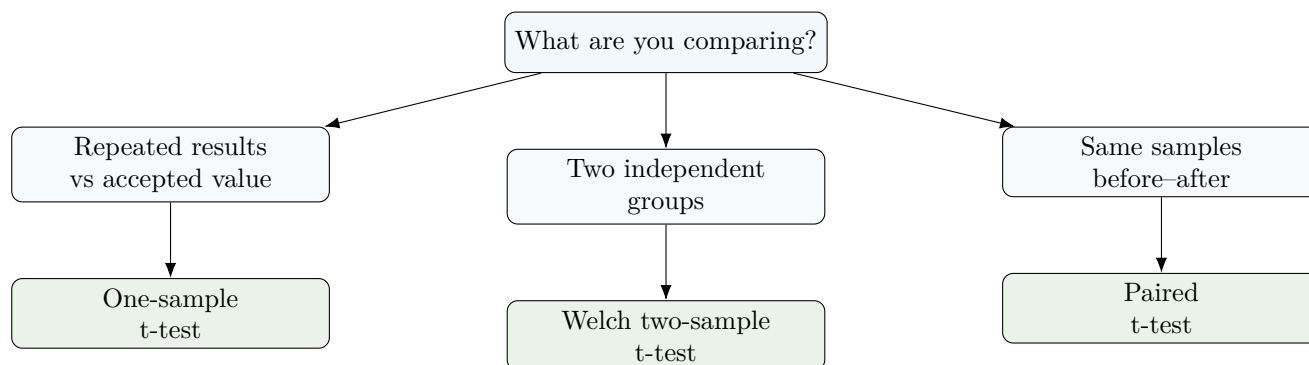
Quick exercise: Choose paired or independent

Decide whether each comparison should use a paired t-test or a two-sample independent t-test.

- The same six samples are tested before and after heating.
- Two different groups of tablets are analysed using two different methods.

- c) Each water sample is split into two portions, and the two portions are analysed by two different methods.

6 Choosing the correct test



Common wrong choices

Do not use a paired t-test just because two columns have the same number of values. Use a paired test only when each value in one column is naturally matched to one value in the other column.

Quick exercise: Test selection

Select the most appropriate test.

- Four replicate titres are compared with a certified concentration.
- Copper concentration is measured by colorimetry in five samples and by titration in five different samples.
- Five samples are measured by colorimetry, then the same five are measured by atomic absorption spectroscopy.

7 Statistical significance vs chemical importance

Two separate questions

A hypothesis test answers a statistical question: is the observed difference too large to explain easily by random scatter? It does not automatically answer the chemical question: does this difference matter in the context of the experiment?

A very precise method can detect tiny differences. For example, a difference of $0.002 \text{ mol dm}^{-3}$ may be statistically significant if the measurements are extremely precise. But for a school acid-base titration, that difference may be chemically unimportant. Conversely, a large difference may fail to reach statistical significance if there are too few repeats or the measurements are very scattered.

Result	Sensible interpretation
Not statistically significant	The data do not provide strong evidence of a real difference. This does not prove the values are identical.
Statistically significant but tiny	The difference is probably real, but it may not matter for the purpose of the experiment.
Statistically significant and chemically large	Strong evidence of a real and practically important difference.
Large-looking but not significant	The experiment may have too much scatter or too few repeats.

Quick exercise: Interpret the conclusion

A t-test gives $p = 0.031$ for the difference between two methods. The mean difference is $0.001 \text{ mol dm}^{-3}$, while the required tolerance for the practical is $0.010 \text{ mol dm}^{-3}$. Write a careful conclusion.

8 Using spreadsheets for t-tests

You can calculate t-tests manually, but spreadsheets are useful for checking work. The formulas below are written in a generic spreadsheet style.

Task	Spreadsheet formula idea
Mean	=AVERAGE(range)
Sample standard deviation	=STDEV.S(range)
One-sample t statistic	=(mean-accepted)/(stdev/SQRT(n))
Two-sample p-value	=T.TEST(range1, range2, 2, 3) for a two-tailed Welch test
Paired p-value	=T.TEST(before"; after, 2, 1) in some spreadsheet systems; check local syntax

Do not outsource interpretation

A spreadsheet can calculate a p-value. It cannot decide whether the method was chemically valid, whether the samples were independent, or whether the difference matters.

Quick exercise: Spreadsheet setup

You have two columns: Method A in A2:A6 and Method B in B2:B6. Write a formula idea for a two-tailed Welch t-test comparing them.

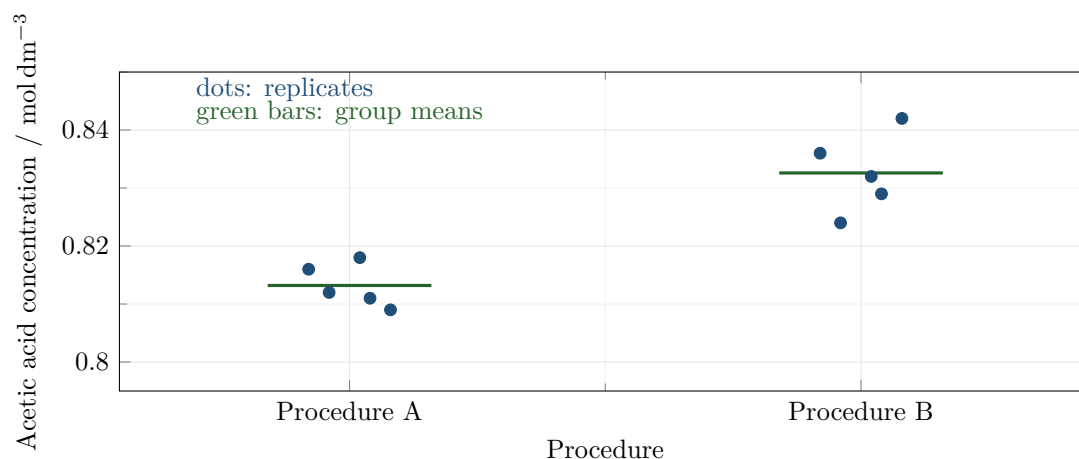
9 Mini-practical: comparing two titration procedures

A class compares two procedures for determining acetic acid concentration in vinegar. Procedure A uses a phenolphthalein endpoint. Procedure B uses a pH probe endpoint. The same vinegar batch is used, but the replicate titrations are performed independently by different groups. The rows below are therefore not matched pairs.

Replicate	Procedure A / mol dm ⁻³	Procedure B / mol dm ⁻³
1	0.812	0.824
2	0.818	0.832
3	0.809	0.842
4	0.816	0.836
5	0.811	0.829

Quick exercise: Analysis task

- 1) State the appropriate test: one-sample, two-sample independent, or paired.
- 2) Calculate the mean and sample standard deviation for each procedure.
- 3) Write H_0 and H_1 .
- 4) Calculate or estimate whether the observed difference is likely to be statistically significant.
- 5) Write a conclusion that includes both statistical and chemical interpretation.



10 Quick reference

Test	Use when	Statistic
One-sample t-test	Repeated results are compared with an accepted value.	$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$
Welch two-sample t-test	Two independent groups are compared.	$t = (\bar{x}_1 - \bar{x}_2) / \sqrt{s_1^2/n_1 + s_2^2/n_2}$
Paired t-test	The same samples are measured twice or naturally matched.	$t = \bar{d} / (s_d / \sqrt{n})$

Degrees of freedom	t_{crit} for two-tailed $\alpha = 0.05$
4	2.776
5	2.571
6	2.447
7	2.365
8	2.306
9	2.262
10	2.228

Answer key

Exercise 1: a) measured standard vs accepted value; b) two analytical methods; c) same samples before and after treatment.

Exercise 2: H_0 : the true mean vitamin C content equals 500 mg. H_1 : the true mean vitamin C content is not equal to 500 mg.

Exercise 3: $t = (50.70 - 50.0)/(0.158/\sqrt{5}) = 9.91$. Since $9.91 > 2.78$, reject H_0 . The method gives a statistically higher value than the accepted concentration.

Exercise 4: $t = (0.812 - 0.835)/\sqrt{0.006^2/4 + 0.009^2/4} = -4.25$ approximately. Since $|t| > 2.45$, the procedures are significantly different by this rough criterion.

Exercise 5: a) paired; b) independent two-sample; c) paired, because each sample is split into matched portions.

Exercise 6: a) one-sample t-test; b) Welch two-sample t-test; c) paired t-test.

Exercise 7: The difference is statistically significant at $\alpha = 0.05$ because $p < 0.05$, but the mean difference is much smaller than the practical tolerance. It is probably not chemically important for this practical.

Exercise 8: Formula idea: =T.TEST(A2:A6,B2:B6,2,3) for a two-tailed Welch test, depending on the spreadsheet syntax.

Exercise 9: Mini-practical: independent two-sample comparison. Means: Procedure A $\bar{x} = 0.8132$, Procedure B $\bar{x} = 0.8326$. Sample standard deviations: A $s = 0.00370$, B $s = 0.00680$. H_0 : the procedure means are equal. H_1 : the procedure means differ. Welch statistic is approximately $t = -5.61$, so the difference is likely statistically significant. Chemically, Procedure B gives a consistently higher concentration; this may indicate endpoint bias, calibration difference, or systematic difference between visual and pH-probe endpoints.